8.6 Solving Logarithmic Equations

A Exponential-Logarithmic Conversion The following two expressions are equivalent: $b^x = y \quad \Leftrightarrow \quad x = \log_b y$ $b > 0, b \neq 1; \quad y > 0, x \in R$	Ex 1. Solve for x. Verify restrictions. a) $\log x = 0$ $\quad x=1$ $x>0$ b) $\ln x = 1$ $\quad x=e$ $x>0$ c) $\log_2(x-1) = 0$ $\quad x-1=1$ $\quad \therefore x=2$ $x>1$ $\quad x=2$ d) $\log(x^2+1) = 1$ $\quad x^2=9$ $\quad \therefore x=+3,-3$ $x^2+1=10$ $\quad x=\pm 3$ e) $\ln(\log x) = 0$ $\quad \ln(\log x)=0$ $\quad \log x = 1$ $x>0$ $\quad \boxed{x=10}$ $\log x>0$ $\boxed{x>1}$
B One-to-one property The logarithmic function is a one-to-one function. So: $\log_b x = \log_b y \quad \Leftrightarrow \quad x = y$ $b > 0, b \neq 1, x \in R, y \in R,$	Ex 2. Solve for x. Verify restrictions. a) $\log(x-1) = \log(2x+1)$ $\quad x>1, x>-\frac{1}{2}$ $\quad \therefore x>1$ $x-1=2x+1$ $-x=2$ $\boxed{x=-2}$ $\rightarrow$ no sol b) $\ln(x+1) - \ln(x-1) = 3$ $\quad \boxed{x>1}$ $x>1$ & >-1 $\ln \frac{x+1}{x-1} = 3$ $\frac{x+1}{x-1} = e^3$ $x+1 = e^3x - e^3$ $x(e^3-1) = e^3+1$ $\therefore \boxed{x = \frac{e^3+1}{e^3-1}}$ c) $\log_2(x-1) + \log_2(x+2) - \log_2(2x-1) = 1$ $\boxed{x>1}$ $\log_2 \frac{(x-1)(x+2)}{(2x-1)} = \log_2 2$ $x^2+x-2 = 4x-2$ $x^2-3x=0$ $x(x-3)=0$ $x=0$ and $x=3$ $\therefore \boxed{x=3}$ d) $\log x = 1 - \log(x-3)$; $x>3$ $\log x(x-3) = \log 10$ $x^2-3x-10=0$ $(x-5)(x+2)=0$ $\quad x=5$, $x=-2$ $\quad \therefore \boxed{x=5}$
C Technology	Ex 3. Use technology (scientific calculator) to find the solution of the following equation to the nearest thousandth. $\ln x + \log x = 5$ $\quad x = e^{\frac{5}{1+\frac{1}{\ln 10}}}$ $\ln x + \frac{\ln x}{\ln 10} = 5$ $\ln x\left(1+\frac{1}{\ln 10}\right) = 5$

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Ex 4. Solve for x.

a) $\log_2(x^2) = (\log_2 x)^2$ $\quad x>0$

$2\log_2 x - (\log_2 x)^2 = 0$

let $y = \log_2 x$

$y^2 - 2y = 0$

$y(y-2) = 0$ → $y=0$, $y=2$

$\log_2 x = 0 \quad x=1$

$\log_2 x = 2 \quad x=4$

$\therefore x = 1, 4$

b) $(\log x)^2 - \log x^2 + 1 = 0$ $\quad x>0$

$y^2 - 2y + 1 = 0$

$(y-1)^2 = 0$

$\log x - 1 = 0$

$\log x = 1$

$\therefore \boxed{x=10}$

c) $\log_{x-1}(4x-4) = 2$ $\quad x>1,\ x\neq 1$

$x-1 \neq 1 \quad x \neq 2$

$\log_{x-1} 4(x-1) = 2$

$\log_{x-1} 4 + 1 = 2$

$\log_{x-1} 4 = 1$

$4 = x-1 \Rightarrow \boxed{x=5}$

d) $\log_2(x-4) + \log_{\sqrt{2}}(x^3-2) + \log_{0.5}(x-4) = 20$ $\quad x>4$

$\log_2(x-4) + \dfrac{\log_2(x^3-2)}{\log_2 \sqrt{2}} + \log_{\frac{1}{2}}(x-4) = 20$

$\log_2(x-4) + 2\log_2(x^3-2) - \log_2(x-4) = 20$

$\log_2(x^3-2) = 10$

$x^3 - 2 = 2^{10}$

$x^3 = 2^{10} + 2$

$x^3 = 1026$

$x = \sqrt[3]{1026}$

e) $4\log\sqrt{x} - 5\sqrt{\log x} = 3$ $\quad x>0$

$2\log x - 5\sqrt{\log x} = 3$

$2\log x - 3 = 5\sqrt{\log x}$

$4\log^2 x - 12\log x + 9 = 25\log x$

$4y^2 - 37y + 9 = 0$

$(4y-1)(y-9) = 0$

$4y - 1 = 0$, $y = \frac{1}{4}$, $x = 10^{\frac{1}{4}}$

$y = 9$, $x = 10^9$

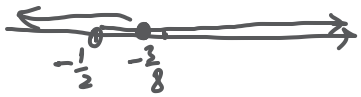
D Inequalities and Logarithms

If $b > 1$ then:

$$\log_b x > \log_b y \quad \Leftrightarrow \quad x > y$$

If $b < 1$ then:

$$\log_b x > \log_b y \quad \Leftrightarrow \quad x < y$$



Ex 5. Solve each inequality.

a) $\log x > 1$ $\quad x>0$

$\log x > \log 10$

$\boxed{x>10}$

b) $\ln(x-1) < 0$ $\quad x>1$

$\ln(x-1) < \ln e$

$x-1 < e$

$x < 1+e$

$\therefore x \in (1, 1+e)$

c) $\log_{0.5}(2x+1) \geq 2$

$x > -\frac{1}{2}$

$\log_{\frac{1}{2}}(2x+1) \geq \log_{\frac{1}{2}}\left(\frac{1}{4}\right)$

$2x+1 \leq \frac{1}{4}$

$2x \leq -\frac{3}{4}$ $\quad x \leq -\frac{3}{8}$

$\therefore x \in \left(-\frac{1}{2}, -\frac{3}{8}\right]$

d) $\log_{0.1} x^2 \leq -1$ $\quad x \neq 0$

$\log_{0.1} x^2 \leq \log_{0.1}(0.1)^{-1}$

$x^2 \geq 10$

$(x-\sqrt{10})(x+\sqrt{10}) \geq 0$ $\quad x \in (-\infty, -\sqrt{10}] \cup [\sqrt{10}, \infty)$

Reading: Nelson Textbook, Pages 487-490
Homework: Nelson Textbook, Page 491: #4acf, 5ace, 6, 7ad, 8, 10, 12, 16, 18, 19, 20

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